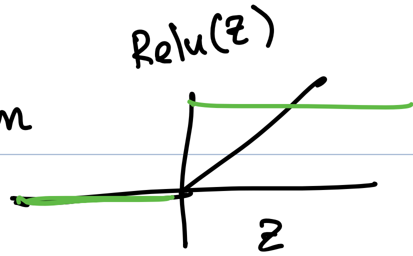
$z = w_1 x + b_1$ preactivation



$h = \mathrm{ReLU}(z)$

$\hat{y} = w_2 h + b_2$ output

Loss $\frac{1}{2}(\hat{y} - y)^2$

Let $x = 2$ $y = 1$ $w_1 = 1$ $b_1 = -1$

$w_2 = 3$ $b_2 = 0$

Forward poss $\boxed{z = 1}$ $\boxed{h = 1}$ $\hat{y} = 3$ $L = 2$

Backward pass

$$\frac{\partial L}{\partial \hat{y}} = (\hat{y} - y) = 2$$

$$\frac{\partial L}{\partial w_2} = \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial w_2} = 2 \cdot h = 2 \cdot 1 = 2$$

$$\frac{\partial L}{\partial b_2} = \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial b_2} = 2 \cdot 1 = 2$$

$$\frac{\partial L}{\partial h} = \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial h} = 2 \cdot w_2 = 2 \cdot 3 = 6$$

$$\frac{\partial L}{\partial z} = \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial h} \cdot \boxed{\frac{\partial h}{\partial z}} = 6 \cdot 1 = 6$$

$$\frac{\partial L}{\partial W_1} = \frac{\partial L}{\partial z} \cdot \frac{\partial z}{\partial W_1} = 6.x = 6.2 = 12$$

$$\frac{\partial L}{\partial b_1} = \frac{\partial L}{\partial z_1} \cdot \frac{\partial z_1}{\partial b_1} = 6.1 = 6$$

Gradient step with learning rate $\eta = \underline{0.1}$

$$W_2 \leftarrow W_2 - \eta_1 \frac{\partial L}{\partial W_2}$$

$$b_2 \leftarrow b_2 - \eta \, \partial L / \partial b_2$$

$$W_1 \leftarrow W_1 - \eta \, \partial L / \partial W_1$$

$$b_1 \leftarrow b_1 - \eta \, \partial L / \partial b_1$$

Recompute $L' = L(y - \hat{y}')$